

Multiple Choice (10 marks)

1. The lunar maria are:
 - (a) ancient heavily cratered highlands
 - (b) dark lavas inside volcanic calderas
 - (c) dark lavas filling older impact basins
 - (d) the bright regions on the Moon
2. What effect or effects would be most significant if the Moon's orbital plane were exactly the same as the ecliptic plane?
 - (a) Solar eclipses would be much rarer.
 - (b) Solar eclipses would last much longer.
 - (c) Solar eclipses would be much more frequent.
 - (d) Solar eclipses would not last as long.
3. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
4. The constellation ... is a bright W-shaped constellation in the northern sky.
 - (a) Centaurus
 - (b) Cygnus
 - (c) Cassiopeia
 - (d) Cepheus
5. Why is the Mars Exploration Rover Spirit currently tilted towards the north?
 - (a) Because it's climbing up a big hill.
 - (b) Because it's in the southern hemisphere where it is winter now.
 - (c) Because it's in the northern hemisphere where it is winter now.
 - (d) Because one of its wheels broke.

True / False (10 marks)

Answer True or False

6. True or False: The Moon's rotational and orbital periods are equal, resulting in the same face always facing the Earth.
7. True or False: Meteorites with high metal content are often fragments of large differentiated asteroids that have been shattered by collisions.
8. True or False: A neutral atom always has equal numbers of neutrons and protons, ensuring stability and balance in the atomic structure.
9. True or False: The telescope was invented by Galileo in 1709, marking the beginning of modern astronomical study.
10. True or False: Meteorites always exhibit a fusion crust due to their passage through the Earth's atmosphere.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of Barnard's Star in the context of stellar proximity to Earth, and analyze its characteristics compared to Proxima Centauri and Alpha Centauri.
12. Discuss the rationale behind Ptolemy's use of 'circles upon circles' in his geocentric model of the universe, particularly in relation to the observed retrograde motion of planets.

End of Paper